



RGPV Bhopal

Information Technology

ACADEMIC PERFORMANCE REPORT

AI-Powered Comprehensive Analytics Report

PROGRAM

B.Tech.

SEMESTER

Semester 3

GENERATED

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Note: Page numbers are approximate and may vary based on data volume. This report contains AI-generated analytics and automatically computed insights.

EXECUTIVE SUMMARY

This report presents a comprehensive analysis of academic performance for 7 students enrolled in the **B.Tech.** program, Semester 3 at **RGPV Bhopal**. The analysis covers overall statistics, branch-wise comparisons, subject-level insights, grade distributions, risk assessments, and AI-generated recommendations for academic improvement.

KEY PERFORMANCE INDICATORS

7 Total Students	83.33% Pass Rate	16.67% Fail Rate	7.17 Avg SGPA
6.99 Avg CGPA	8.13 Highest SGPA	5.71 Lowest SGPA	7.61 Median SGPA

KEY FINDINGS

- **Good academic outcome** — 83.33% pass rate with 1 students requiring support.
- **Batch Topper:** AASTHA AGRAWAL (0105IT241001) achieved the highest SGPA of 8.13.
- The average SGPA of **7.17** is satisfactory but has scope for improvement.
- **Best performing department:** Information Technology.
- **Most challenging subject:** IT305- [T] recorded the highest failure rate.
- **1 student(s)** identified as academically at-risk and require targeted intervention.

PERFORMANCE SNAPSHOT

Metric	Value
Total Students Evaluated	7
Students Passed	5 (83.33%)
Students Failed	1 (16.67%)
Average SGPA	7.17
Median SGPA	7.61

Metric	Value
Standard Deviation	1.05
Average CGPA	6.99
Highest SGPA	8.13 (AASTHA AGRAWAL)
Lowest SGPA	5.71
Departments Analyzed	1
Subjects Analyzed	12

OVERALL STATISTICS

Complete statistical overview of 7 students across 1 department(s) and 12 subject(s).

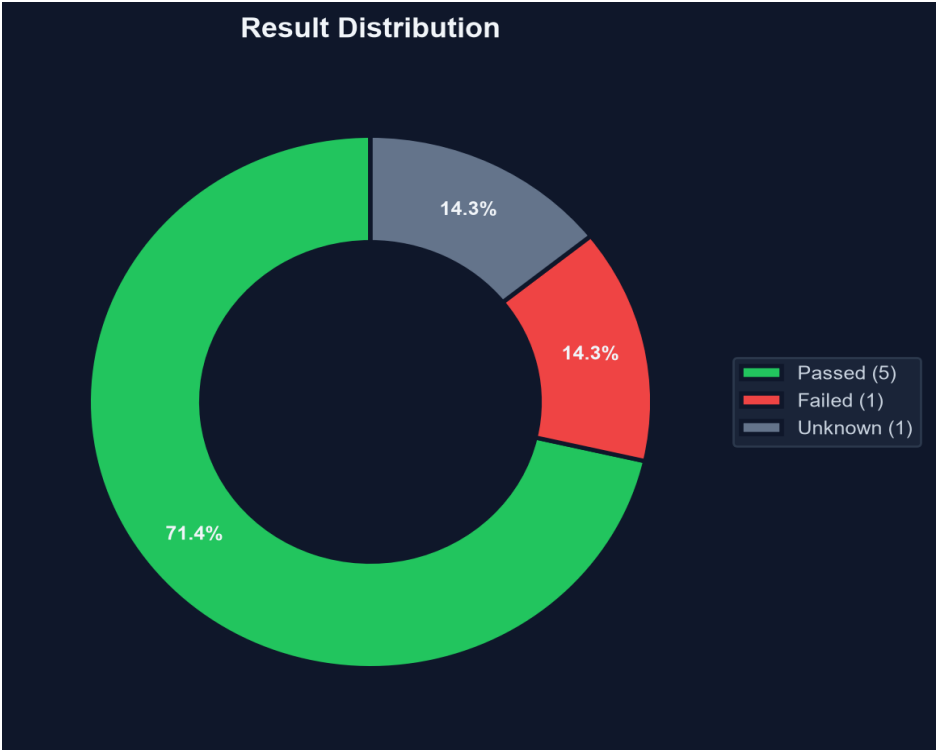
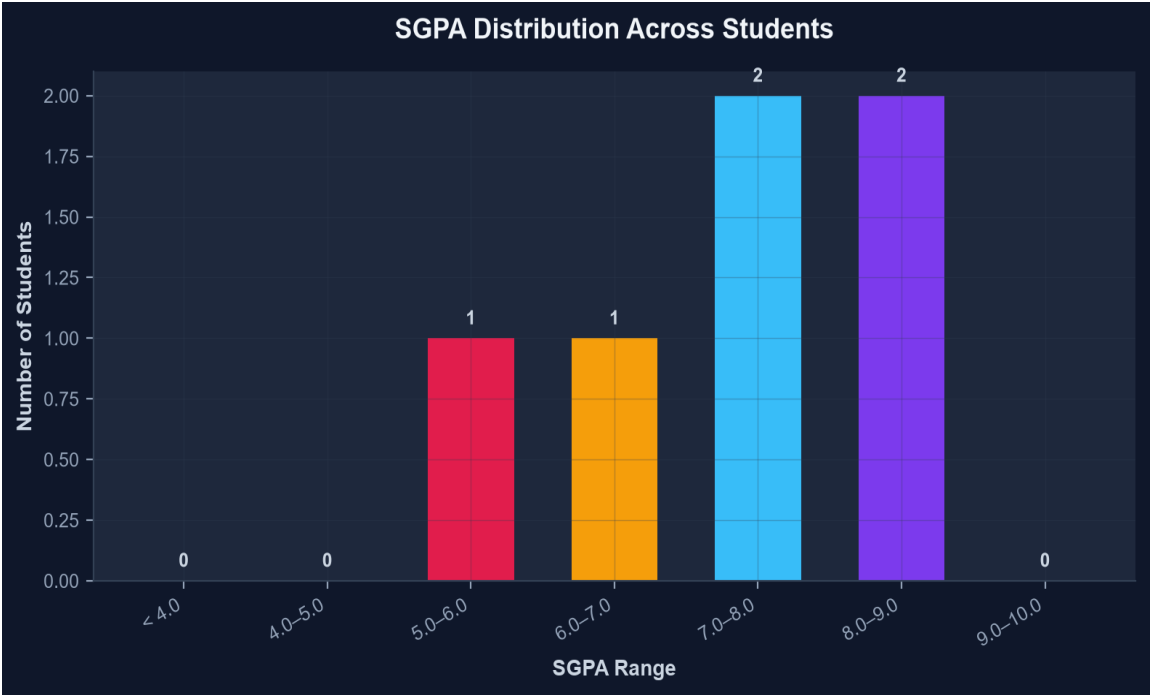
RESULT BREAKDOWN

Category	Count	Percentage
Total Students	7	100%
Passed	5	83.33%
Failed	1	16.67%
Unknown/Pending	1	14.3%

SGPA STATISTICS

Statistic	SGPA	CGPA
Highest	8.13	8.18
Lowest	5.71	5.64
Average (Mean)	7.17	6.99
Median	7.61	7.25
Std. Deviation	1.05	—

SGPA DISTRIBUTION



BRANCH-WISE ANALYSIS

Comparative performance analysis across 1 academic department(s). This section evaluates each branch based on average SGPA, pass rate, failure rate, and backlog distribution.

DEPARTMENT COMPARISON TABLE

Branch	Students	Pass %	Fail %	Avg SGPA	Avg CGPA	Backlogs	Topper
IT	7	83.33%	16.67%	7.17	6.99	1	AASTHA AGRAWAL

SEMESTER ANALYSIS

Performance analysis for **Semester 3** of the **B.Tech.** program. This section provides a snapshot of the semester's academic outcomes.

SEMESTER PERFORMANCE OVERVIEW

Metric	Value
Semester	Semester 3
Program	B.Tech.
Total Students	7
Pass Count	5
Fail Count	1
Pass Percentage	83.33%
Average SGPA	7.17
Average CGPA	6.99
Highest SGPA	8.13
Lowest SGPA	5.71
Number of Subjects	12
Departments	1

DEPARTMENT-WISE SEMESTER RESULTS

Department	Students	Pass %	Avg SGPA	Topper
Information Technology	7	83.33%	7.17	AASTHA AGRAWAL

Note: Multi-semester trend analysis requires comparison data from previous semesters. Use the --compare flag with a previous semester's Excel file to enable trend charts.

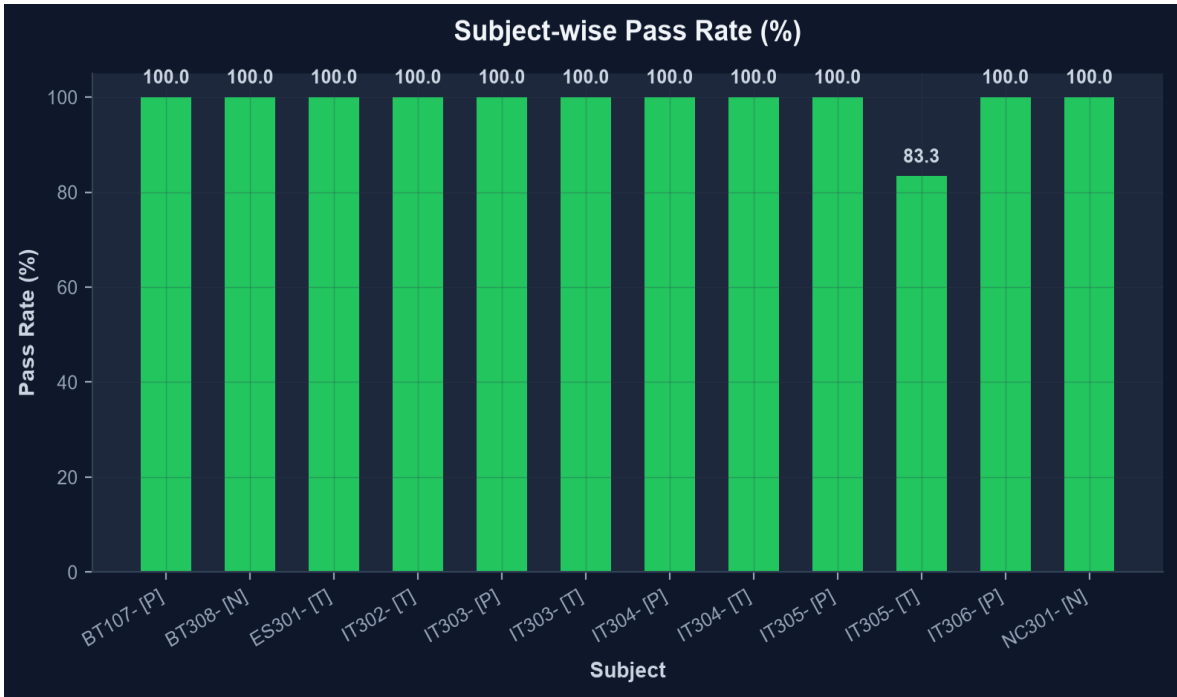
SUBJECT-WISE ANALYSIS

Detailed analysis of **12** subject(s) evaluated in this semester. Each subject is assessed for pass rate, failure rate, average grade, difficulty index, and grade distribution.

SUBJECT PERFORMANCE SUMMARY

Subject	Students	Pass %	Fail %	Avg GP	Difficulty	Highest	Lowest
IT305- [T]	6	83.33%	16.67%	4.0	0.6	B+	F
BT107- [P]	6	100.0%	0.0%	8.0	0.2	A+	B
BT308- [N]	6	100.0%	0.0%	0.0	1.0	—	—
ES301- [T]	6	100.0%	0.0%	6.17	0.38	B+	C
IT302- [T]	6	100.0%	0.0%	4.17	0.58	B+	C
IT303- [P]	6	100.0%	0.0%	8.5	0.15	A+	B+
IT303- [T]	6	100.0%	0.0%	4.67	0.53	B	C
IT304- [P]	6	100.0%	0.0%	8.33	0.17	A+	B+
IT304- [T]	6	100.0%	0.0%	3.83	0.62	B+	C
IT305- [P]	6	100.0%	0.0%	8.5	0.15	A+	B+
IT306- [P]	6	100.0%	0.0%	8.33	0.17	A+	B+
NC301- [N]	6	100.0%	0.0%	0.0	1.0	—	—

SUBJECT-WISE PASS RATE



SUBJECT DIFFICULTY INDEX

The Difficulty Index ranges from 0.0 (easiest) to 1.0 (hardest), calculated based on average grade points. Higher values indicate more challenging subjects.



DETAILED SUBJECT BREAKDOWN

- **BT107- [P]:** 6 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 8.0, Failed: 0 — Grades: [A: 3, A+: 2, B: 1]
- **BT308- [N]:** 6 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 0.0, Failed: 0 — Grades: [NA: 6]
- **ES301- [T]:** 6 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 6.17, Failed: 0 — Grades: [B: 3, B+: 2, C: 1]
- **IT302- [T]:** 6 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 4.17, Failed: 0 — Grades: [B: 1, B+: 2, C: 1, C+: 2]
- **IT303- [P]:** 6 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 8.5, Failed: 0 — Grades: [A: 1, A+: 4, B+: 1]
- **IT303- [T]:** 6 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 4.67, Failed: 0 — Grades: [B: 3, C: 2, C+: 1]
- **IT304- [P]:** 6 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 8.33, Failed: 0 — Grades: [A: 2, A+: 3, B+: 1]
- **IT304- [T]:** 6 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 3.83, Failed: 0 — Grades: [B: 1, B+: 1, C: 2, C+: 2]
- **IT305- [P]:** 6 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 8.5, Failed: 0 — Grades: [A: 1, A+: 4, B+: 1]

- **IT305- [T]**: 6 students, Pass Rate: 83.33%, Fail Rate: 16.67%, Avg Grade Point: 4.0, Failed: 1 — Grades: [B: 2, B+: 1, C: 1, D##: 1, F: 1]
- **IT306- [P]**: 6 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 8.33, Failed: 0 — Grades: [A: 2, A+: 3, B+: 1]
- **NC301- [N]**: 6 students, Pass Rate: 100.0%, Fail Rate: 0.0%, Avg Grade Point: 0.0, Failed: 0 — Grades: [NA: 6]

■ **Most Failed Subject: IT305- [T]**

✓ **Easiest Subject: IT303- [P] (Difficulty Index: 0.15)**

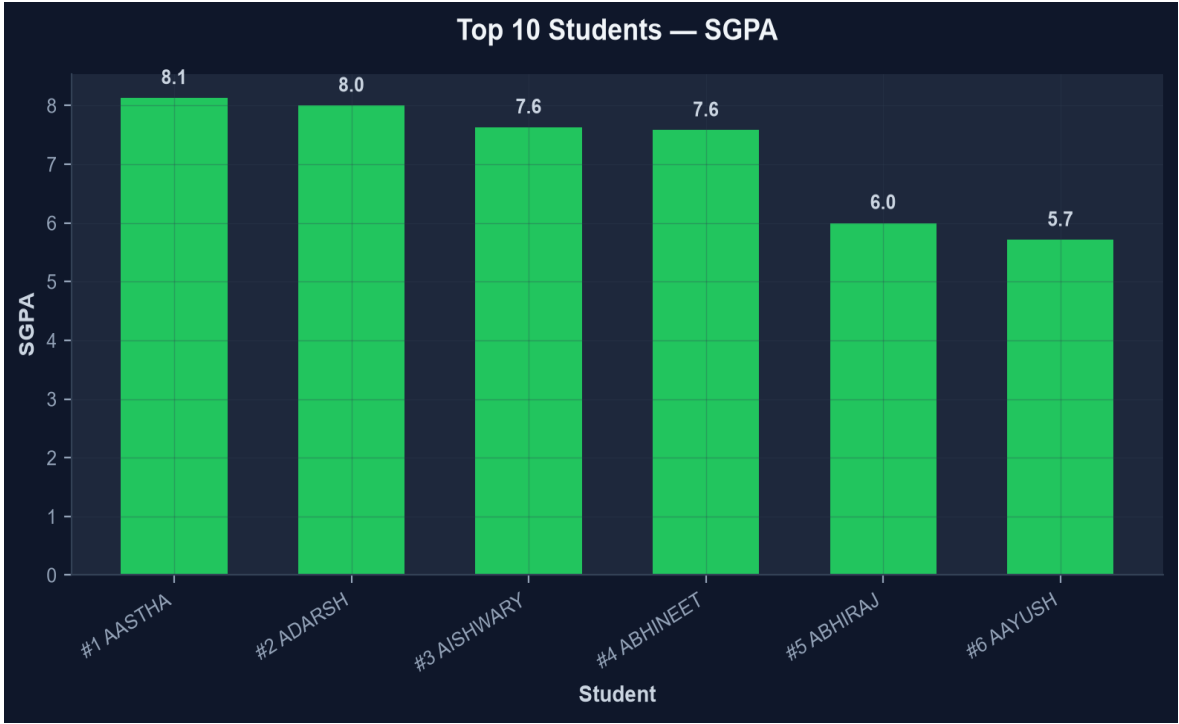
TOPPER ANALYSIS

Academic rank list of top-performing students. Rankings are based on Semester GPA (SGPA), with CGPA as a tiebreaker.

TOP 10 STUDENTS

Rank	Enrollment	Name	Branch	SGPA	CGPA	Result
■	0105IT241001	AASTHA AGRAWAL	IT	8.13	8.18	PASS
■	0105IT241005	ADARSH KUMAR DWIVEDI	IT	8.0	7.43	PASS
■	0105IT241007	AISHWARYA MOHAN DAS	IT	7.63	7.67	PASS
4	0105IT241003	ABHINEET BHARDWAJ	IT	7.58	7.07	PASS
5	0105IT241004	ABHIRAJ SIROTHIYA	IT	6.0	5.94	PASS
6	0105IT241002	AAYUSH MEHRA	IT	5.71	5.64	FAIL

TOP STUDENTS — SGPA VISUALIZATION



STUDENT PERFORMANCE DETAILS

Individual performance cards for 7 student(s). Each card shows subject grades, SGPA, CGPA, result status, and backlog count.

#1 AASTHA AGRAWAL0105IT241001

Branch: Information Technology | Program: B.Tech. | Semester: 3

SUBJECT CODE	GRADE
ES301- [T]	B+
IT302- [T]	B+
IT303- [T]	B
IT304- [T]	B+
IT305- [T]	B
BT107- [P]	A
IT303- [P]	A+
IT304- [P]	A
IT305- [P]	A+
IT306- [P]	A
BT308- [N]	NA
NC301- [N]	NA

SGPA

8.13

CGPA

8.18

RESULT

PASS

BACKLOGS

0

#2 ADARSH KUMAR DWIVEDI

0105IT241005

Branch: Information Technology | Program: B.Tech. | Semester: 3

SUBJECT CODE	GRADE
ES301- [T]	B+
IT302- [T]	C+
IT303- [T]	B
IT304- [T]	B
IT305- [T]	B+
BT107- [P]	A+
IT303- [P]	A+
IT304- [P]	A+
IT305- [P]	A+
IT306- [P]	A+
BT308- [N]	NA
NC301- [N]	NA

SGPA

8.0

CGPA

7.43

RESULT

PASS

BACKLOGS

0



#3 AISHWARYA MOHAN DAS

0105IT241007

Branch: Information Technology | Program: B.Tech. | Semester: 3

SUBJECT CODE	GRADE
ES301- [T]	B
IT302- [T]	B
IT303- [T]	C+
IT304- [T]	C+
IT305- [T]	B
BT107- [P]	A+
IT303- [P]	A+
IT304- [P]	A+
IT305- [P]	A+
IT306- [P]	A+
BT308- [N]	NA
NC301- [N]	NA

SGPA

7.63

CGPA

7.67

RESULT

PASS

BACKLOGS

0



#4 ABHINEET BHARDWAJ

0105IT241003

Branch: Information Technology | Program: B.Tech. | Semester: 3

SUBJECT CODE	GRADE
ES301- [T]	B
IT302- [T]	B+
IT303- [T]	B
IT304- [T]	C+
IT305- [T]	C
BT107- [P]	A
IT303- [P]	A+
IT304- [P]	A+
IT305- [P]	A+
IT306- [P]	A+
BT308- [N]	NA
NC301- [N]	NA

SGPA

7.58

CGPA

7.07

RESULT

PASS

BACKLOGS

0



#5 ABHIRAJ SIROTHIYA

0105IT241004

Branch: Information Technology | Program: B.Tech. | Semester: 3

SUBJECT CODE	GRADE
ES301- [T]	B
IT302- [T]	C
IT303- [T]	C
IT304- [T]	C
IT305- [T]	D##
BT107- [P]	B
IT303- [P]	B+
IT304- [P]	B+
IT305- [P]	B+
IT306- [P]	B+
BT308- [N]	NA
NC301- [N]	NA

SGPA

6.0

CGPA

5.94

RESULT

PASS

BACKLOGS

0



#6 AAYUSH MEHRA

0105IT241002

Branch: Information Technology | Program: B.Tech. | Semester: 3

SUBJECT CODE	GRADE
ES301- [T]	C
IT302- [T]	C+
IT303- [T]	C
IT304- [T]	C
IT305- [T]	F
BT107- [P]	A
IT303- [P]	A
IT304- [P]	A
IT305- [P]	A
IT306- [P]	A
BT308- [N]	NA
NC301- [N]	NA

SGPA

5.71

CGPA

5.64

RESULT

FAIL

BACKLOGS

1



#7 0105IT241006

0105IT241006

Branch: Information Technology | Program: B.Tech. | Semester: 3

SUBJECT CODE

GRADE

SGPA

—

CGPA

—

RESULT

UNKNOWN

BACKLOGS

0

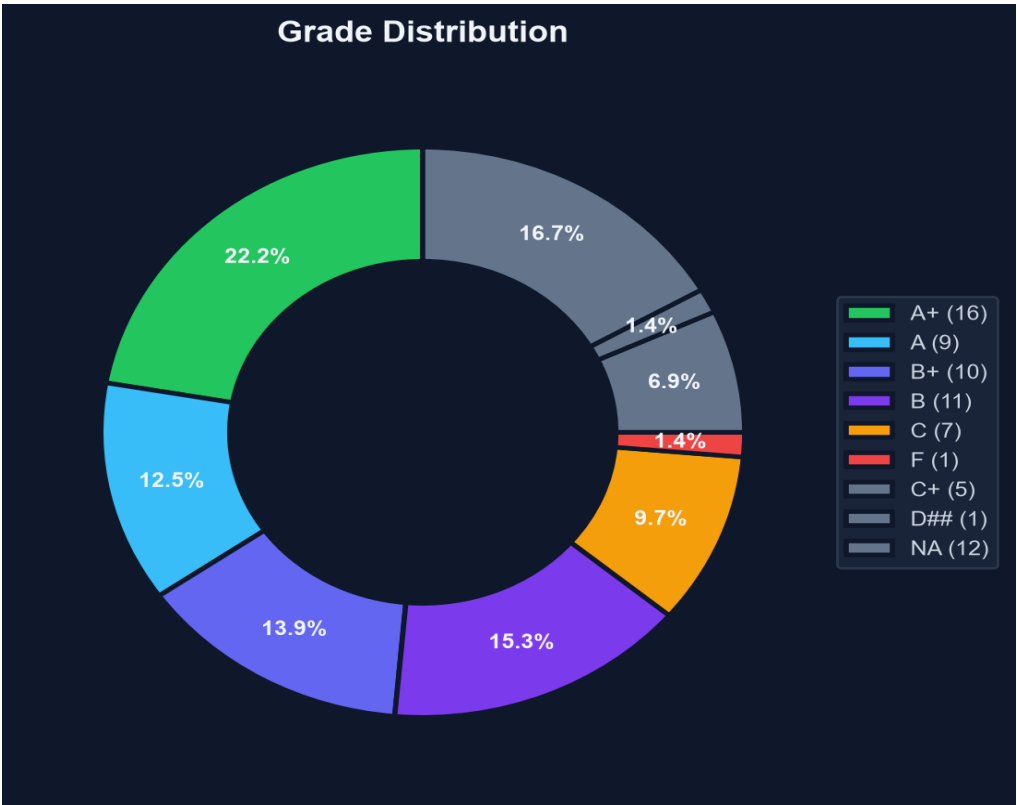
GRADE DISTRIBUTION

Distribution analysis of 72 total grades awarded across all subjects and students.

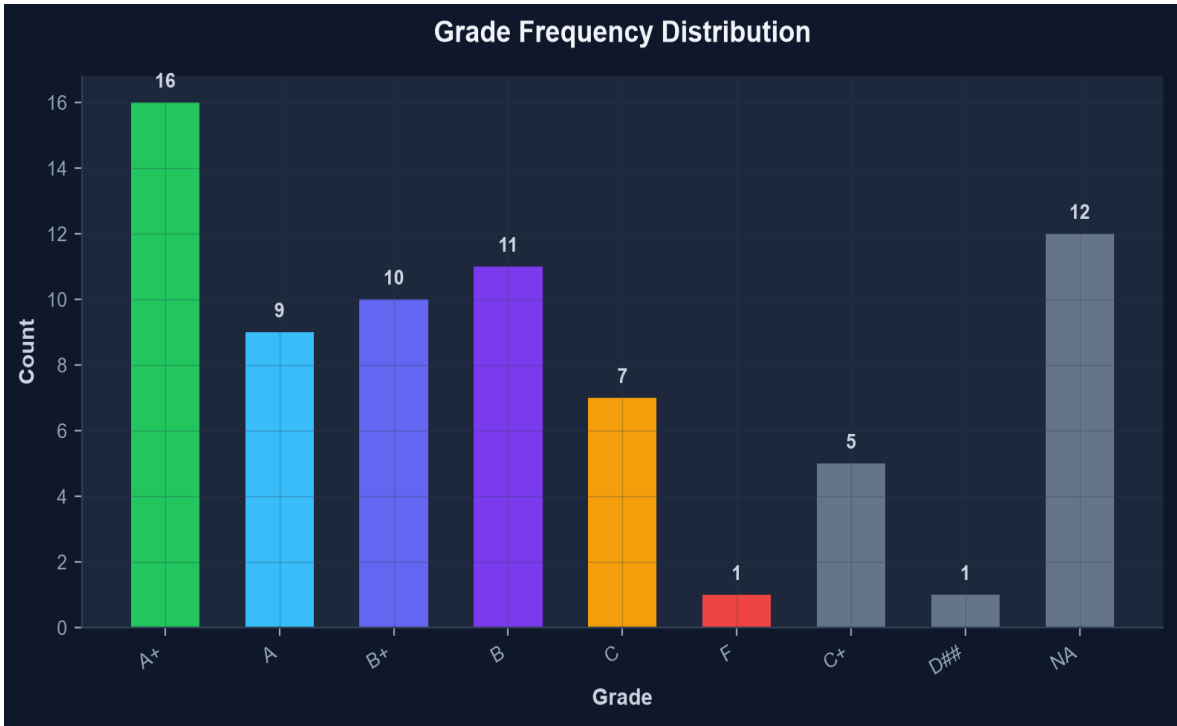
GRADE FREQUENCY TABLE

Grade	Count	Percentage	Performance Level
A+	16	22.2%	Excellent
A	9	12.5%	Very Good
B+	10	13.9%	Good
B	11	15.3%	Above Average
C	7	9.7%	Average
F	1	1.4%	Fail
C+	5	6.9%	C+
D##	1	1.4%	D##
NA	12	16.7%	NA

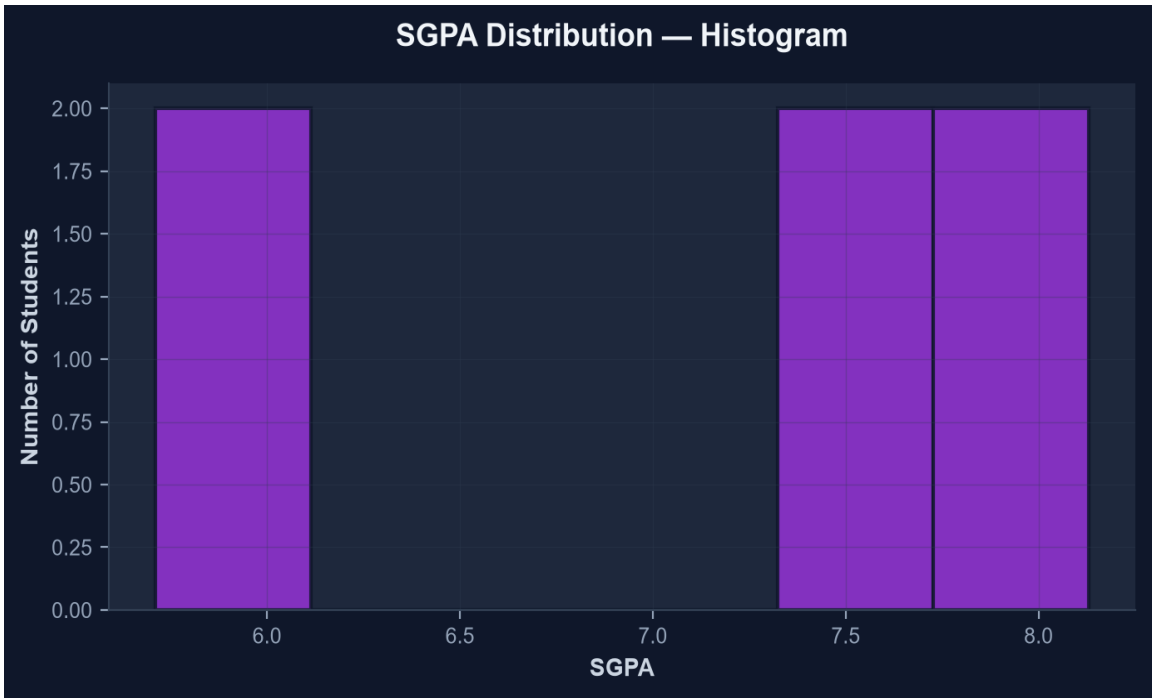
GRADE DISTRIBUTION — DONUT CHART



GRADE DISTRIBUTION — BAR CHART



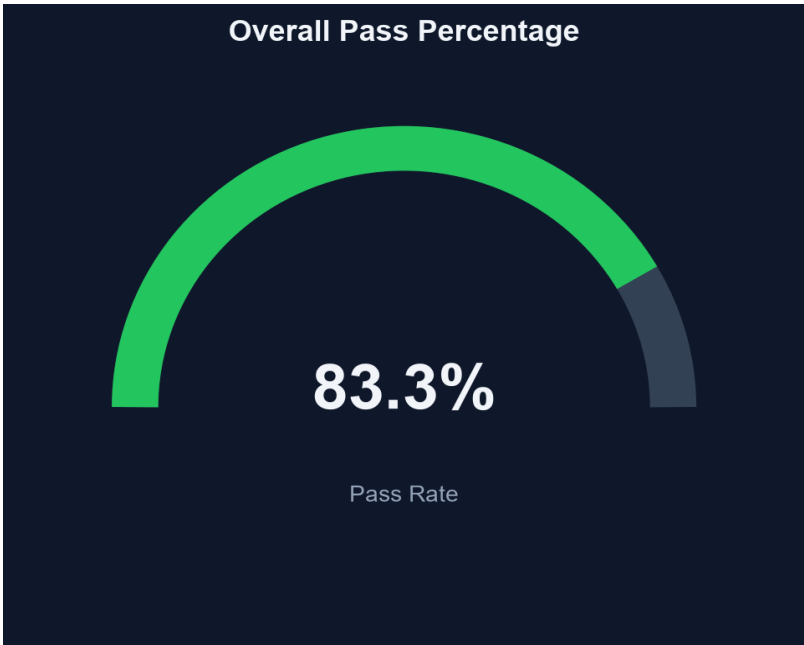
SGPA FREQUENCY HISTOGRAM



PASS PERCENTAGE ANALYSIS

Overall pass rate analysis showing the proportion of students who cleared all subjects versus those with one or more failures.

OVERALL PASS RATE GAUGE



DEPARTMENT-WISE PASS RATE

Department	Total	Passed	Failed	Pass %
Information Technology	7	5	1	83.33%



BACKLOG ANALYSIS

Analysis of 1 total backlog instances across 1 student(s). Backlogs include F, Fail, AB, Absent, and RL grades.

DEPARTMENT-WISE BACKLOG DISTRIBUTION

Department	Total Backlogs
Information Technology	1



SUBJECT-WISE BACKLOG DISTRIBUTION

Subject Code	Failure Count
IT305	1



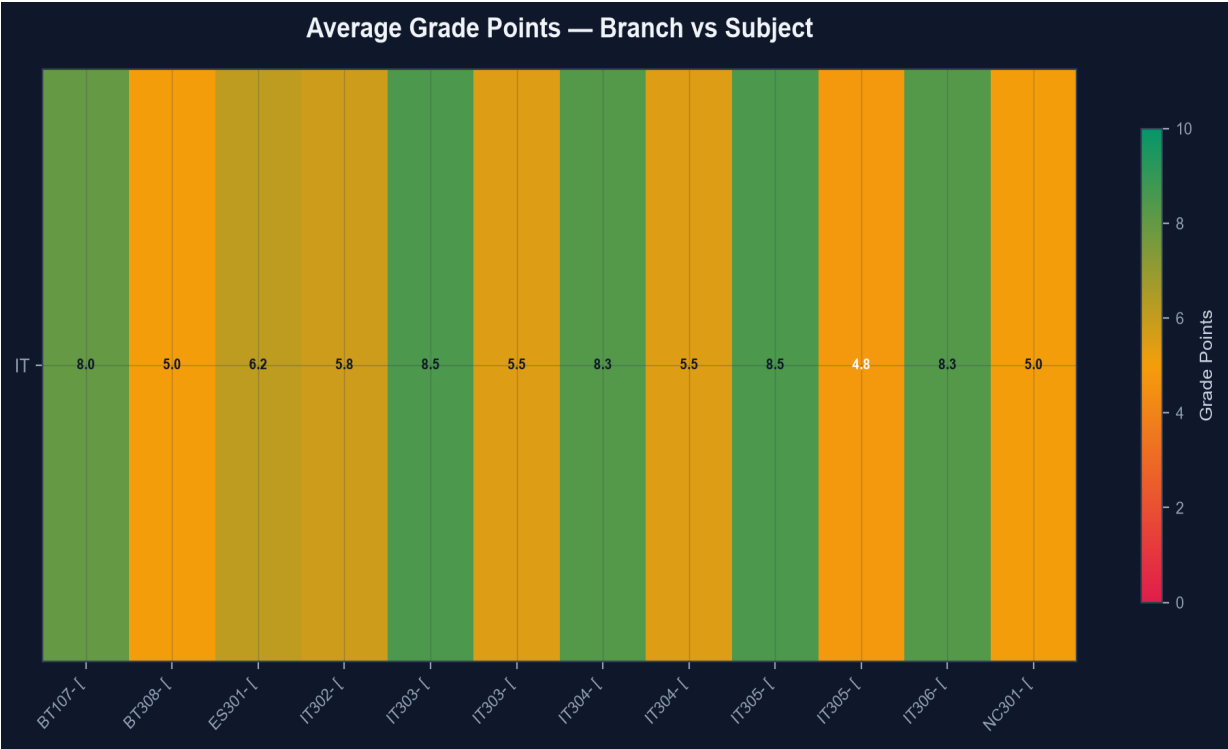
STUDENTS WITH BACKLOGS

Enrollment	Name	Branch	SGPA	Backlogs	Failed Subjects
0105IT241002	AAYUSH MEHRA	IT	5.71	1	IT305

PERFORMANCE HEATMAPS

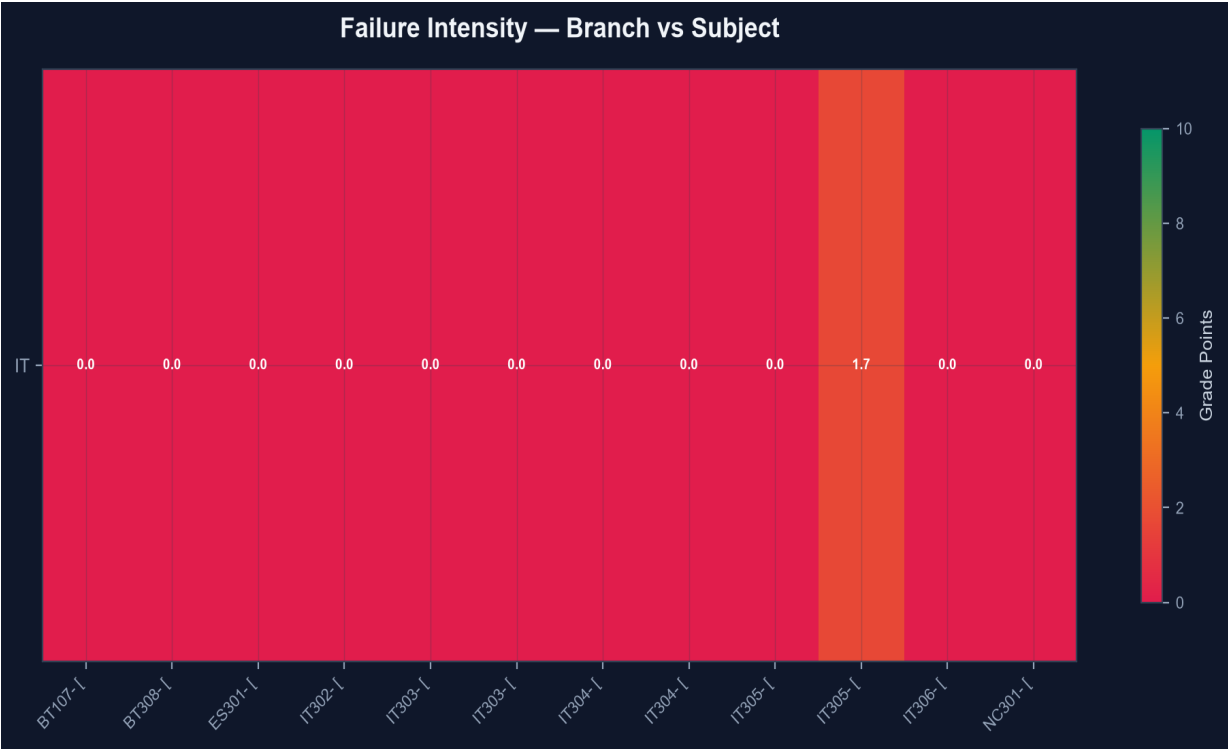
Visual heatmaps showing performance intensity across branches and subjects. Warmer colors (red/orange) indicate weaker performance; cooler colors (green) indicate stronger results. Grade points range from 0 (Fail) to 10 (Outstanding).

BRANCH × SUBJECT PERFORMANCE HEATMAP



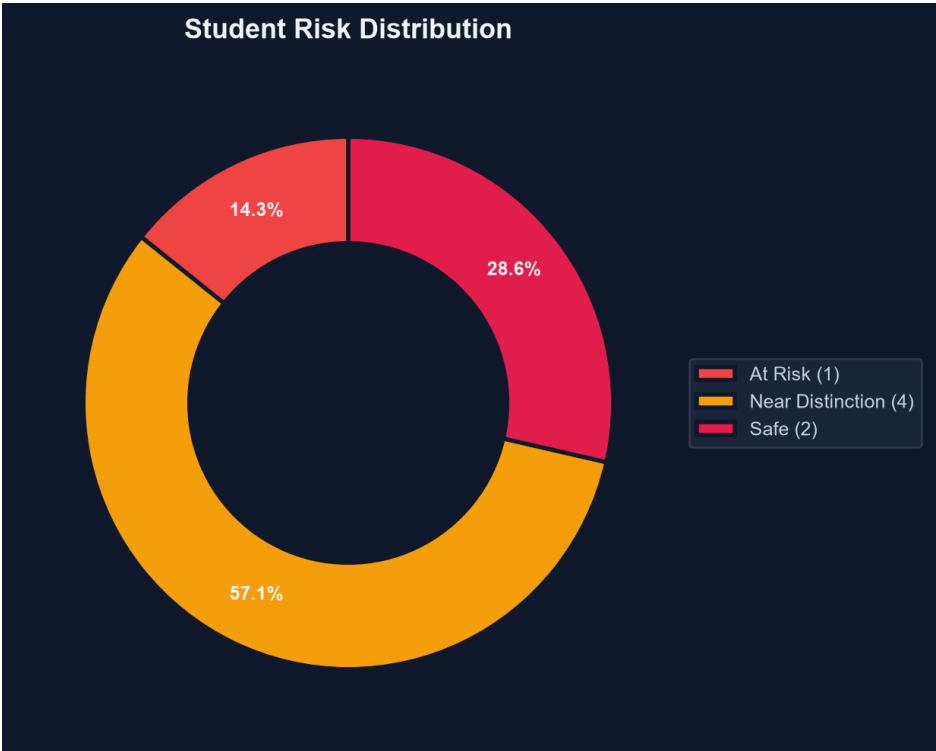
FAILURE INTENSITY HEATMAP

Higher values (red) indicate higher failure rates. Scale: 0 = No Failures, 10 = 100% Failure.



RISK ANALYSIS

Academic risk stratification identifying students who need attention. **1** at-risk, **4** near-distinction, **0** with multiple backlogs.



STUDENTS AT RISK (SGPA < 5.0 or FAIL status)

Enrollment	Name	Branch	SGPA	CGPA	Backlogs
0105IT241002	AAYUSH MEHRA	Information Tec	5.71	5.64	1

STUDENTS NEAR DISTINCTION (SGPA 7.5–8.5)

Enrollment	Name	Branch	SGPA	CGPA
0105IT241001	AASTHA AGRAWAL	Information Tec	8.13	8.18
0105IT241005	ADARSH KUMAR DWIVE	Information Tec	8.0	7.43
0105IT241007	AISHWARYA MOHAN DA	Information Tec	7.63	7.67
0105IT241003	ABHINEET BHARDWAJ	Information Tec	7.58	7.07

IMPROVEMENT ANALYSIS

Improvement analysis requires comparison data from a previous semester or job. To enable this section, generate the report with the **--compare** flag pointing to a previous result Excel file.

Example: `python -m report_generator.generate_report --input current.xlsx --compare previous.xlsx`

COMPARISON ANALYSIS

Comparison analysis is available when a previous dataset is provided for reference. Use the **--compare** flag to specify a previous result Excel file.

Example: `python -m report_generator.generate_report --input current.xlsx --compare previous.xlsx`

AI-GENERATED INSIGHTS

The AI analytics engine has generated **42** observations based on comprehensive statistical analysis of the academic data. These insights are automatically derived from patterns, distributions, and trends.

- 1

A total of 7 students were analyzed in this academic evaluation.
- 2

Overall pass percentage stands at 83.33%, with 5 students clearing all subjects.
- 3

Failure rate is recorded at 16.67%, affecting 1 students.
- 4

Good overall performance with pass rate above 75%, though room for improvement exists.
- 5

Average SGPA across all students is 7.17 out of 10.0.
- 6

Median SGPA is 7.61, indicating the central tendency of student performance.
- 7

Median SGPA exceeds average by 0.44 points, indicating some very low scores are dragging the average down.
- 8

Highest SGPA achieved is 8.13 by AASTHA AGRAWAL (0105IT241001).
- 9

Lowest SGPA recorded is 5.71.
- 10

Standard deviation of SGPA is 1.05, measuring the spread of academic performance.
- 11

Average Cumulative GPA (CGPA) is 6.99.
- 12

Information Technology is the best-performing department based on average SGPA.
- 13

Information Technology is the weakest-performing department and may need academic support.
- 14

Information Technology department: 7 students, 83.33% pass rate, average SGPA 7.17.
- 15

Information Technology recorded 1 total subject backlogs.
- 16

Most failed subject: IT305- [T] — requires focused remedial action.
- 17

Subject BT107- [P] achieved 100% pass rate across all 6 students.
- 18

Subject BT107- [P] has an excellent average grade point of 8.0.

-
- 19 Subject BT308- [N] achieved 100% pass rate across all 6 students.
 - 20 Subject BT308- [N] has a high difficulty index of 1.0 (scale 0-1).
-

- 21 Subject ES301- [T] achieved 100% pass rate across all 6 students.
- 22 Subject IT302- [T] achieved 100% pass rate across all 6 students.
- 23 Subject IT303- [P] achieved 100% pass rate across all 6 students.
- 24 Subject IT303- [P] has an excellent average grade point of 8.5.
- 25 Subject IT303- [T] achieved 100% pass rate across all 6 students.
- 26 Subject IT304- [P] achieved 100% pass rate across all 6 students.
- 27 Subject IT304- [P] has an excellent average grade point of 8.33.
- 28 Subject IT304- [T] achieved 100% pass rate across all 6 students.
- 29 Subject IT304- [T] has a high difficulty index of 0.62 (scale 0-1).
- 30 Subject IT305- [P] achieved 100% pass rate across all 6 students.
- 31 Subject IT305- [P] has an excellent average grade point of 8.5.
- 32 Subject IT306- [P] achieved 100% pass rate across all 6 students.
- 33 Subject IT306- [P] has an excellent average grade point of 8.33.
- 34 Subject NC301- [N] achieved 100% pass rate across all 6 students.
- 35 Subject NC301- [N] has a high difficulty index of 1.0 (scale 0-1).
- 36 1 student(s) identified as academically at-risk (SGPA < 5.0 or FAIL status).
- 37 4 student(s) are near distinction level (SGPA 7.5–8.5) — targeted support could push them higher.
- 38 'A+' grade was the achieved 16 times (22.2% of all grades).
- 39 1 'F' (fail) grades recorded (1.4% of all grades), each representing a subject backlog.
- 40 2 students (28.6%) scored SGPA above 8.0, demonstrating strong academic excellence.

-
- 41 Batch topper is AASTHA AGRAWAL (0105IT241001) with SGPA 8.13.
-
- 42 Top 3 students maintain an average SGPA of 7.92.
-

AI RECOMMENDATIONS

The AI engine has generated 17 actionable recommendations based on the analysis findings. Recommendations are prioritized as **HIGH**, **MEDIUM**, and **LOW** priority.

Priority Distribution: 1 HIGH | 8 MEDIUM | 8 LOW

HIGH	R1	Identify and assign academic mentors to 1 at-risk student(s) immediately.
MEDIUM	R2	Review curriculum difficulty for BT308- [N] (difficulty index: 1.0). Consider adjusting teaching methodology or evaluation pattern.
MEDIUM	R3	Review curriculum difficulty for NC301- [N] (difficulty index: 1.0). Consider adjusting teaching methodology or evaluation pattern.
MEDIUM	R4	Schedule one-on-one counseling sessions with all at-risk students to understand personal/academic challenges.
MEDIUM	R5	Provide targeted coaching to 4 near-distinction student(s) — small improvement could push them to distinction level.
MEDIUM	R6	Increase practical and hands-on laboratory sessions to improve conceptual understanding.
MEDIUM	R7	Implement continuous internal assessment to provide early warning signals for academic difficulties.
MEDIUM	R8	Organize industry expert guest lectures to improve student engagement and practical understanding.
MEDIUM	R9	Implement a digital learning management system for supplementary resources and self-paced revision.
LOW	R10	Recognize and reward top-performing students with merit certificates and academic incentives.
LOW	R11	Encourage toppers to participate in peer tutoring programs to uplift overall batch performance.
LOW	R12	Conduct student feedback surveys on teaching quality and curriculum relevance.

LOW

R13 Establish study groups and collaborative learning environments for improved peer learning.

LOW

R14 Review and update course materials to align with current industry standards and practices.

LOW

R15 Organize academic orientation sessions at the start of each semester to set performance expectations.

LOW

R16 Publish this analytics report to department heads and academic coordinators for data-driven planning.

LOW

R17 Schedule regular parent-teacher meetings for students with declining performance trends.